Polynomials — problem 10.4

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Given a monic polynomial $f(x)$ of degree n over $\mathbb{Z}$ and $k, p \in \mathbb{N}$, prove that if none of the numbers $f(k), f(k+1), \ldots, f(k+p)$ is divisible by $p+1$, then $f(x) = 0$ has no rational solution.